

Configuring an AWS Transit Gateway

To Begin with the Lab

Summary of the Lab

In this lab, an **AWS Transit Gateway** was configured to connect three VPCs within the same AWS Region. The environment included one VPC with both public and private subnets and two additional VPCs with private subnets. An Internet Gateway was attached to the first VPC to provide internet access for a Jump Host, while private EC2 instances were deployed in each VPC. Each VPC was attached to the Transit Gateway, and route tables were updated to direct inter-VPC traffic through it. Finally, connectivity tests confirmed that all private EC2 instances could communicate successfully, demonstrating centralized, scalable, and efficient networking using AWS Transit Gateway.

Note: This lab assumes that three VPCs have already been created. **VPC-A** contains one public subnet (for the Jump Host) and one private subnet, while **VPC-B** and **VPC-C** each contain one private subnet. The EC2 instances and security groups are already configured. The objective of this lab is to connect all three VPCs using an **AWS Transit Gateway**.

- Create 3 VPCs (same Region, e.g., us-east-1): **cf-vpc-A** (10.0.0.0/16), **cf-vpc-B** (10.1.0.0/16), **cf-vpc-C** (10.2.0.0/16).

Your VPCs

VPCs | VPC encryption controls

Your VPCs (3) Info Last updated 9 minutes ago Actions Create VPC

Find VPCs by attribute or tag

Name = cf-vpc-A X Name = cf-vpc-B X Name = cf-vpc-C X Clear filters

☐	Name	VPC ID	State	Encr...	Encryption control ...	Block Public...	IPv4 CIDR
☐	cf-vpc-A	vpc-01a6a3e3b5712e37e	Available.	-	-	Off	10.0.0.0/16
☐	cf-vpc-B	vpc-0ded0ba4d08f9898a	Available.	-	-	Off	10.1.0.0/16
☐	cf-vpc-C	vpc-065ac0e918fd69d07	Available.	-	-	Off	10.2.0.0/16

Select a VPC above

- Create Subnets: **cf-vpc-A** → 1 Public (10.0.1.0/24) + 1 Private (10.0.2.0/24); **cf-vpc-B** → 1 Private (10.1.1.0/24); **cf-vpc-C** → 1 Private (10.2.1.0/24).

Subnets (4) Info Last updated 1 minute ago Actions Create subnet

Find subnets by attribute or tag

VPC = vpc-065ac0e918fd69d07 X VPC = vpc-0ded0ba4d08f9898a X VPC = vpc-01a6a3e3b5712e37e X Clear filters

☐	Name	Subn...	State	VPC	Bloc...	IPv4 C...	IPv6 ...	IPv6 ...	Avail...	Availability Zone
☐	cf-private-subnet-C	subnet...	Available.	vpc-065ac0e...	Off	10.2.1.0/24	-	-	251	use1-az4 (us-east-1a)
☐	cf-private-subnet-B	subnet...	Available.	vpc-0ded0ba...	Off	10.1.1.0/24	-	-	251	use1-az4 (us-east-1a)
☐	cf-private-subnet-A	subnet...	Available.	vpc-01a6a3e...	Off	10.0.2.0/24	-	-	251	use1-az4 (us-east-1a)
☐	cf-public-subnet-A	subnet...	Available.	vpc-01a6a3e...	Off	10.0.1.0/24	-	-	251	use1-az4 (us-east-1a)

- Create an Internet Gateway (IGW): Attach it only to **cf-vpc-A**.

Internet gateways (1/1) [Info](#)

Find internet gateways by attribute or tag

VPC ID = vpc-01a6a3e3b5712e37e [X](#) [Clear filters](#)

< 1 > [Settings](#)

☒	Name	Internet gateway ID	State	VPC ID	Owner
☒	cf-igw-a	igw-0bbba2785ae84999e	Attached	vpc-01a6a3e3b5712e37e cf-vpc-A	463646775279

- Create Route Tables: Public RT (cf-vpc-A) → 0.0.0.0/0 → IGW; Private RTs → only local route initially.

Route tables (4) [Info](#)

Last updated 1 minute ago [Actions](#) [Create route table](#)

Find route tables by attribute or tag

VPC = vpc-01a6a3e3b5712e37e [X](#) VPC = vpc-065ac0e918fd69d07 [X](#) VPC = vpc-0ded0ba4d08f9898a [X](#) [Show more \(+1\)](#) [Clear filters](#)

< 1 > [Settings](#)

☐	Name	Route...	Explicit subnet associations	Edge...	Main	VPC	Own...
☐	cf-public-rt-a	rtb-05e...	subnet-05f4093344701926f / cf-public-subnet-A	-	No	vpc-01a6a3e3b5712e37e cf-vpc-A	463646...
☐	cf-private-rt-c	rtb-040...	subnet-090ea5e9282ceb13e / cf-private-subnet-C	-	No	vpc-065ac0e918fd69d07 cf-vpc-C	463646...
☐	cf-private-rt-b	rtb-0ec...	subnet-0e514d699d8259bbd / cf-private-subnet-B	-	No	vpc-0ded0ba4d08f9898a cf-vpc-B	463646...
☐	cf-private-rt-a	rtb-0a4...	subnet-0e6c32039a3b0a1c2 / cf-private-subnet-A	-	No	vpc-01a6a3e3b5712e37e cf-vpc-A	463646...

- Routes of cf-public-rt-a:

rtb-05ecbd58172d2d030 / cf-public-rt-a [Settings](#) [Dropdown](#)

[Details](#) [Routes](#) [Subnet associations](#) [Edge associations](#) [Route propagation](#) [Tags](#)

Routes (2) [Both](#) [Edit routes](#)

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0bbba2785ae84999e	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

- Launch a total of **four EC2 instances**: one **Jump Host (Bastion Host)** in the **public subnet of cf-vpc-A** for SSH access

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name: Bastion Host [Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

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Recents My AMIs [Quick Start](#)

Amazon Linux

macOS

Ubuntu

Windows

Red Hat

SUSE Linux

Debian

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.18 AMI
ami-06067086cf86c58e6 (64-bit (x86), uefi-preferred) / ami-071bc399fae13263b (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs [Free tier eligible](#)

Summary

Number of instances [Info](#): 1

Software Image (AMI)
Amazon Linux 2023 AMI 2023.12...[read more](#)
ami-06067086cf86c58e6

Virtual server type (instance type)
t3.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

EC2 > Instances > Launch an instance

Network settings

VPC - required

vpc-01a6a3e3b5712e37e (cf-vpc-A)

Subnet

subnet-05f4093344701926f cf-public-subnet-A

Auto-assign public IP

Enable

Summary

Number of instances: 1

Software Image (AMI)

Amazon Linux 2023 AMI 2023.12...read more

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel Launch instance Preview code

Firewall (security groups)

Create security group Select existing security group

Security group name - required

cf-vpc-a-sg

Description - required

cf-vpc-a-sg

Inbound Security Group Rules

Security group rule 1 (TCP: 22, 0.0.0.0/0)

- three private EC2 instances—one each in the private subnets of cf-vpc-A, cf-vpc-B, and cf-vpc-C—to test connectivity through the Transit Gateway.

Instances (1/4)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone
cf-ec2-private-b	i-0b6d1349bf222cc45	Running	t2.micro	Initializing	us-east-1a
cf-ec2-private-a	i-0fe7bac7ef1d78025	Running	t3.micro	3/3 checks passed	us-east-1a
cf-ec2-private-c	i-0cbba930dc560a06a	Pending	t2.micro	-	us-east-1a
Bastion Host	i-02124e5b2427e93ee	Running	t3.micro	3/3 checks passed	us-east-1a

- Navigate to **Transit Gateways** in the VPC console.
- Click **Create Transit Gateway**.

VPC > Transit gateways

You can visualize and monitor your Transit Gateway(s) from the AWS Network Manager. Register your Transit Gateway by creating a global network to get started.

Transit gateways

Create transit gateway

Name	Transit gateway ID	Owner ID	State
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- Provide a meaningful name such as **cf-vpc-a-b-c**.

VPC > Transit gateways > Create transit gateway

Create transit gateway

A transit gateway (TGW) is a network transit hub that interconnects attachments (VPCs and VPNs) within the same AWS account or across AWS accounts.

Details - optional

Name tag

cf-vpc-a-b-c-tgw

Description

description

- Leave all other settings as their default values.
- Click **Create Transit Gateway** and wait until its status changes to **Available**.

You successfully created tgw-08b8bd886cb7b99b7 / cf-vpc-a-b-c-tgw.

You can visualize and monitor your Transit Gateway(s) from the [AWS Network Manager](#). Register your Transit Gateway by creating a [global network](#) to get started.

Transit gateways (1) [Info](#)

Find transit gateway by attribute or tag

☐	Name ↗	Transit gateway ID	Owner ID	State
☐	cf-vpc-a-b-c-tgw	tgw-08b8bd886cb7b99b7	463646775...	Pending

- Open the **Transit Gateway Attachments** page.
- Click **Create Transit Gateway Attachment**.

VPC > Transit gateway attachments

You can visualize and monitor your Transit Gateway(s) from the [AWS Network Manager](#). Register your Transit Gateway by creating a [global network](#) to get started.

Transit gateway attachments [Info](#)

Find transit gateway attachment by attribute or tag

☐	Name ↗	Transit gateway attachment ID	Transit gateway ID	State	Resource ty...	Resource ID
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- Select the Transit Gateway that was created in the previous step.
- Choose **VPC** as the attachment type.

Create transit gateway attachment [Info](#)

A transit gateway (TGW) is a network transit hub that interconnects attachments (VPCs and VPNs) within the same AWS account or across AWS accounts.

Details

Name tag - optional
Creates a tag with the key set to Name and the value set to the specified string.

cf-tgw-attachment

Transit gateway ID [Info](#)
tgw-08b8bd886cb7b99b7

Attachment type [Info](#)
VPC

- Select **cf-vpc-A**.
- Choose the **Private Subnet** of **cf-vpc-A** for the attachment.
- Click **Create Attachment**.

VPC attachment
Select and configure your VPC attachment.

☒ DNS support [Info](#)

☒ Security Group Referencing support [Info](#)

☐ IPv6 support [Info](#)

☐ Appliance Mode support [Info](#)

VPC ID
Select the VPC to attach to the transit gateway.

vpc-01a6a3e3b5712e37e

Subnet IDs [Info](#)
Select the subnets in which to create the transit gateway VPC attachment.

☒ us-east-1a [↗](#) subnet-0e6c32039a3b0a1c2

☐ us-east-1b No subnet available

- Repeat the same process for **cf-vpc-B** and **cf-vpc-C**.

You successfully created VPC attachment tgw-attach-013fdf2d026503b3a / cf-tgw-attachment-c.

Transit gateway attachments (3) [Info](#)

Find transit gateway attachment by attribute or tag

☐	Name ↗	Transit ...	Transit ...	State	Resource ty...	Resource ID	Association route table ID
☐	cf-tgw-attachment-c	tgw-attach...	tgw-08b8b...	Pending	VPC	vpc-065ac0e918fd69d07	--
☐	cf-tgw-attachment-b	tgw-attach...	tgw-08b8b...	Pending	VPC	vpc-0ded0ba4d08f9898a	--
☐	cf-tgw-attachment-a	tgw-attach...	tgw-08b8b...	Available	VPC	vpc-01a6a3e3b5712e37e	tgw-rtb-04a14c0ece1a004d2

- Open the **Transit Gateway Route Tables** section.
- Select the default route table that was automatically created with the Transit Gateway.
- Open the **Associations**, **Propagations**, and **Routes** tabs.
- Verify that all three VPC attachments are automatically associated and that the CIDR blocks of all three VPCs have been propagated into the Transit Gateway Route Table.

Transit gateway route tables: tgw-rtb-04a14c0ece1a004d2

Details | **Associations** | Propagations | Prefix list references | Routes | Tags

Associations (3) Info

Find association by attribute or tag

☐	Attachment ID	Resource type	Resource ID	State
☐	tgw-attach-0bb4f4c5ab7fce72d	VPC	vpc-0ded0ba4d08f9898a	✔ Associated
☐	tgw-attach-0d7b1bdd707d31fbb	VPC	vpc-01a6a3e3bd5712e37e	✔ Associated
☐	tgw-attach-013fdfd026503b3a	VPC	vpc-065ac0e918fd69d07	✔ Associated

- Open the **Route Table** associated with the **Private Subnet** of **cf-vpc-A**, **cf-vpc-B** and **cf-vpc-C**.
- Click **Edit Routes**.
- Add a new route with the destination **10.0.0.0/8**.
- Select the **Transit Gateway** as the target.
- Save the route.

VPC > Route tables > rtb-05259d319c8ee3dd6 > Edit routes

Edit routes

Destination	Target	Status	Propagated	Route Origin
10.1.0.0/16	local	✔ Active	No	CreateRouteTable
10.0.0.0/8	Transit Gateway	-	No	CreateRoute

Buttons: Add route, Remove, Cancel, Preview, Save changes

- Same for cf-private-rt-b

rtb-0ec499d5ff159e46f / cf-private-rt-b

Details | **Routes** | Subnet associations | Edge associations | Route propagation | Tags

Routes (2)

Filter routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/8	tgw-08b8bd886cb7b99b7	✔ Active	No	Create Route
10.1.0.0/16	local	✔ Active	No	Create Route Table

- And cf-private-rt-c

rtb-040e7946bdc39901 / cf-private-rt-c



Details Routes Subnet associations Edge associations Route propagation Tags

Routes (2)

Both

Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
10.0.0.0/8	tgw-08b8bd886cb7b99b7	Active	No	Create Route
10.2.0.0/16	local	Active	No	Create Route Table

Do the same thing for the **cf-public-rt-a**

rtb-05ecbd58172d2d030 / cf-public-rt-a



Details Routes Subnet associations Edge associations Route propagation Tags

Routes (3)

Both

Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-0bbba2785ae84999e	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table
10.0.0.0/8	tgw-08b8bd886cb7b99b7	Active	No	Create Route

Let's check the traffic for all the instance

This is the traffic for **cf-ec2-private-a**, **cf-ec2-private-b** and **cf-ec2-private-c**

```
[ec2-user@ip-10-0-1-160 ~]$ ping 10.0.2.254
PING 10.0.2.254 (10.0.2.254) 56(84) bytes of data:
64 bytes from 10.0.2.254: icmp_seq=1 ttl=127 time=0.164 ms
64 bytes from 10.0.2.254: icmp_seq=2 ttl=127 time=0.203 ms
64 bytes from 10.0.2.254: icmp_seq=3 ttl=127 time=0.176 ms
^C
--- 10.0.2.254 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2095ms
rtt min/avg/max/mdev = 0.164/0.181/0.203/0.016 ms
[ec2-user@ip-10-0-1-160 ~]$
[ec2-user@ip-10-0-1-160 ~]$
[ec2-user@ip-10-0-1-160 ~]$
[ec2-user@ip-10-0-1-160 ~]$ ping 10.1.1.205
PING 10.1.1.205 (10.1.1.205) 56(84) bytes of data:
64 bytes from 10.1.1.205: icmp_seq=1 ttl=126 time=3.05 ms
64 bytes from 10.1.1.205: icmp_seq=2 ttl=126 time=0.849 ms
64 bytes from 10.1.1.205: icmp_seq=3 ttl=126 time=0.841 ms
^C
--- 10.1.1.205 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2059ms
rtt min/avg/max/mdev = 0.841/1.578/3.045/1.037 ms
[ec2-user@ip-10-0-1-160 ~]$
[ec2-user@ip-10-0-1-160 ~]$
[ec2-user@ip-10-0-1-160 ~]$
[ec2-user@ip-10-0-1-160 ~]$ ping 10.2.1.252
PING 10.2.1.252 (10.2.1.252) 56(84) bytes of data:
64 bytes from 10.2.1.252: icmp_seq=1 ttl=126 time=1.17 ms
64 bytes from 10.2.1.252: icmp_seq=2 ttl=126 time=0.696 ms
64 bytes from 10.2.1.252: icmp_seq=3 ttl=126 time=0.688 ms
64 bytes from 10.2.1.252: icmp_seq=4 ttl=126 time=0.782 ms
64 bytes from 10.2.1.252: icmp_seq=5 ttl=126 time=0.666 ms
^C
--- 10.2.1.252 ping statistics ---
5 packets transmitted, 5 received, 0% packet loss, time 4127ms
rtt min/avg/max/mdev = 0.666/0.799/1.165/0.187 ms
[ec2-user@ip-10-0-1-160 ~]$
```